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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2018 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Friday 17th August 2018

General Instructions

- Reading time — 5 minutes
- Writing time — 2 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference Sheet
- Candidature — 122 boys

Examiner

WJM

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

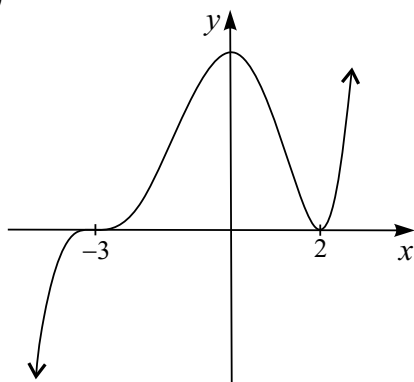
The point R divides the interval from $A(-4, 1)$ to $B(2, 7)$ internally in the ratio $5 : 1$. What are the coordinates of R ?

- (A) $(6, 1)$
- (B) $(1, 6)$
- (C) $(-3, 2)$
- (D) $(2, -3)$

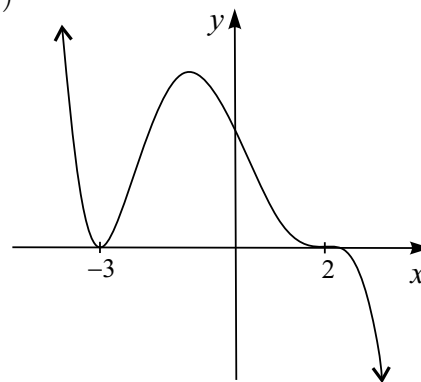
QUESTION TWO

Which of the following diagrams could represent the graph of $y = (x + 3)^2(2 - x)^3$?

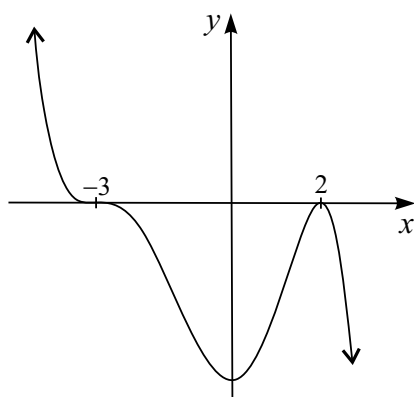
(A)



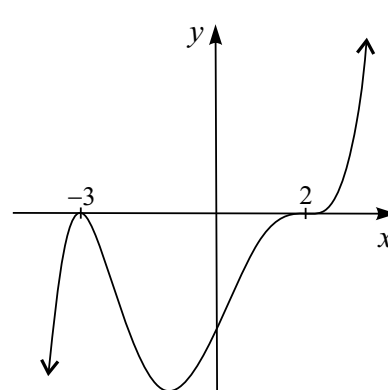
(C)



(B)



(D)



QUESTION THREE

What is the acute angle between the lines $3x + y + 2 = 0$ and $y = 2 + x$, correct to the nearest degree?

- (A) 64°
- (B) 27°
- (C) 45°
- (D) 63°

QUESTION FOUR

What is the derivative of $\sin^{-1} 2x$?

- (A) $\frac{2}{\sqrt{1-4x^2}}$
- (B) $\frac{1}{\sqrt{4-x^2}}$
- (C) $\frac{-2}{\sqrt{1-4x^2}}$
- (D) $\frac{2}{1+4x^2}$

QUESTION FIVE

A particle is moving along the x -axis. Its velocity $v \text{ ms}^{-1}$ is given by $v = \sqrt{6x - x^3}$. What is the acceleration of the particle when $x = 2$?

- (A) 2 ms^{-2}
- (B) $-\frac{3}{2} \text{ ms}^{-2}$
- (C) -6 ms^{-2}
- (D) -3 ms^{-2}

QUESTION SIX

What is the value of k given that $\int_0^k \frac{dx}{9+x^2} = \frac{\pi}{9}$?

(A) $\frac{1}{\sqrt{3}}$

(B) $\sqrt{3}$

(C) $3\sqrt{3}$

(D) 3

QUESTION SEVEN

A function is defined by $f(x) = (x+2)^2 + 1$ for $x \leq -2$. What is the inverse function of $f(x)$?

(A) $f^{-1}(x) = -2 \pm \sqrt{x-1}$

(B) $f^{-1}(x) = -2 + \sqrt{x-1}$

(C) $f^{-1}(x) = -2 - \sqrt{x-1}$

(D) $f^{-1}(x) = \frac{1}{(x+2)^2 + 1}$

QUESTION EIGHT

A parabola can be represented by the parametric equations $x = 6t$, $y = -3t^2 + 3$. What are the coordinates of the focus?

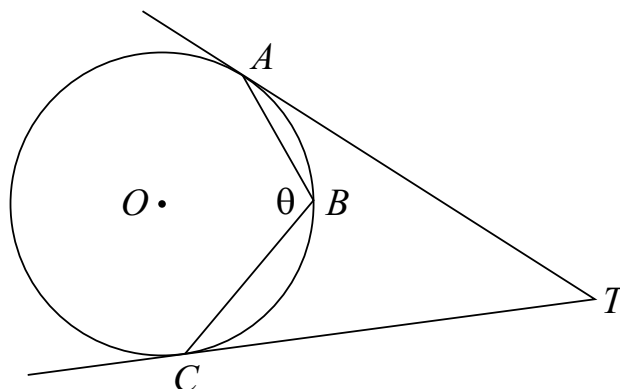
(A) (0, 3)

(B) (0, 0)

(C) (0, -3)

(D) (0, -6)

QUESTION NINE



The diagram above shows a circle through A , B and C , with centre O . Tangents at A and C intersect at T , and $\angle ABC = \theta$.

What is the size of $\angle ATC$ in terms of θ ?

- (A) $2\theta - 180^\circ$
- (B) $180^\circ - \theta$
- (C) $\theta - 90^\circ$
- (D) $2\theta - 90^\circ$

QUESTION TEN

A polynomial is defined by $P(x) = ax^4 + 2bx^3 + 4cx^2 + 8dx + 16e$ for constants a , b , c , d and e . It is known that $x - 2$ is a factor of $P(x)$, and when $P(x)$ is divided by $x + 2$ the remainder is 32.

What is the value of $b + d$?

- (A) 1
- (B) -1
- (C) 16
- (D) -16

_____ End of Section I _____

Examination continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.	Marks
(a) Factorise $8x^3 - y^3$.	1
(b) Find $\int \frac{dx}{\sqrt{9-x^2}}$.	1
(c) Find $\int \cos^2 x \, dx$.	2
(d) The equation $3x^3 - 18x^2 - 25x + 15 = 0$ has roots α , β and γ .	
(i) Write down the values of $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$.	1
(ii) Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$.	1
(e) Solve the inequation $\frac{6}{x+2} \leq 1$.	3
(f) Evaluate $\int_0^{\frac{\pi}{2}} \cos x (1 - \sin x)^2 \, dx$ using the substitution $u = 1 - \sin x$.	3
(g) Find the term independent of x in the expansion of $\left(3x^2 - \frac{2}{x}\right)^9$.	3

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) (i) State the domain and range of $y = \sin^{-1}(x + 1) + \frac{\pi}{2}$. 2

(ii) Hence sketch the graph of $y = \sin^{-1}(x + 1) + \frac{\pi}{2}$. 1

(b) (i) Show that $\operatorname{cosec} 2\alpha - \cot 2\alpha = \tan \alpha$. 2

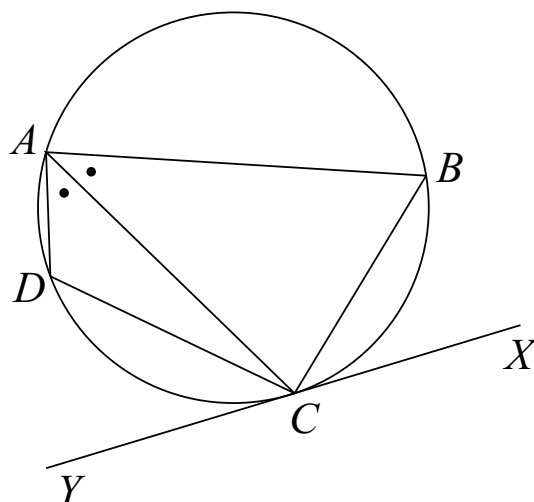
(ii) Hence find the exact value of $\tan 15^\circ$. 1

(c) A continuous function is defined by $f(x) = \tan \frac{x}{2} - x$ for $-\pi < x < \pi$.

(i) Show that $f(x)$ has a root between $x = 2$ and $x = 3$. 1

(ii) Taking $x = 2.5$ as an initial approximation, use Newton's method once to find a better approximation of the root. Give the value of your approximation correct to two decimal places. 2

(d) 2

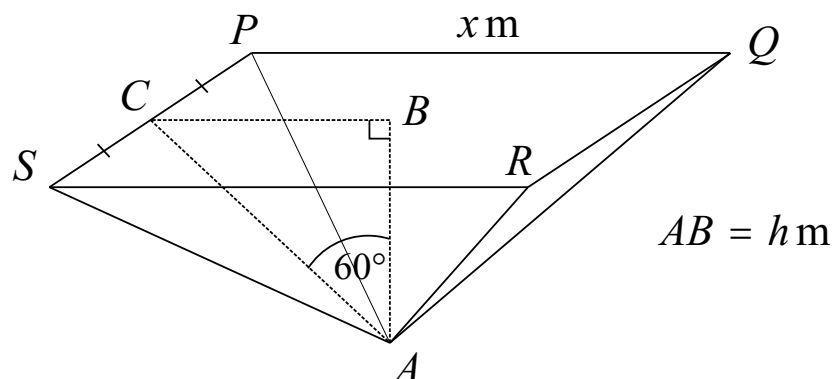


The diagram above shows the cyclic quadrilateral $ABCD$, such that AC bisects $\angle BAD$. The line XY is a tangent to the circle at C .

Show that BD is parallel to XY .

QUESTION TWELVE (Continued)

(e)

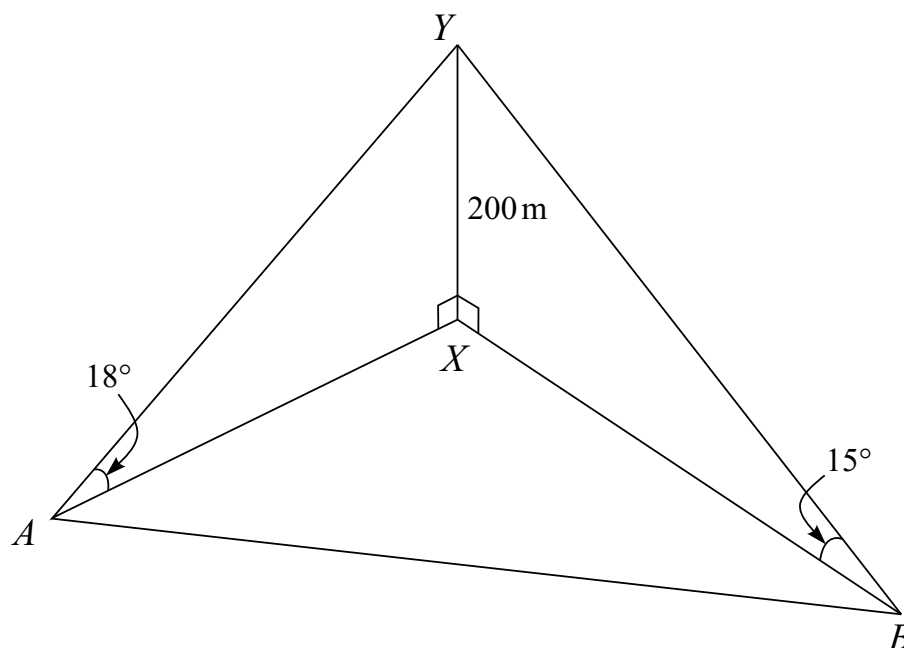


The diagram above shows a square pyramid with base $PQRS$. The centre of the base is the point B , and the perpendicular height AB of the pyramid is h m. The side length of the square base is x m, and the slant height AC of the pyramid makes an angle of 60° to the vertical, as shown in the diagram.

- (i) Show that $x = 2h\sqrt{3}$. 1
- (ii) Hence show that $h = \left(\frac{V}{4}\right)^{\frac{1}{3}}$, where V is the volume of the pyramid in cubic metres. 1
- (iii) The pyramid is initially empty, and is filled with water at a constant rate of $0.5 \text{ m}^3/\text{min}$. Find that rate at which the height of the water is increasing after it has been filled for 8 minutes. 2

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



Abby and Bob are bushwalking on flat ground. They can both see the same mountain, which has a height of 200 m. From Abby's position A , the mountain has a bearing of 035°T , and the top of the mountain has an angle of elevation of 18° . From Bob's position B , the mountain has a bearing of 335°T , and the top of the mountain has an angle of elevation of 15° . Let the base and the top of the mountain be the points X and Y respectively.

(i) Show that $\angle AXB = 60^\circ$.

1

(ii) Find the distance from Abby to Bob. Give your answer correct to the nearest metre.

2

(b) The roots of $x^3 + 9x^2 + kx - 216 = 0$ form a geometric progression. Find the roots.

3

(c) A particle moves such that its displacement in metres after t seconds is given by $x = \sqrt{3}\sin 2t - \cos 2t - 1$.

(i) Prove that the particle is moving in simple harmonic motion by showing that $\ddot{x} = -4(x + 1)$.

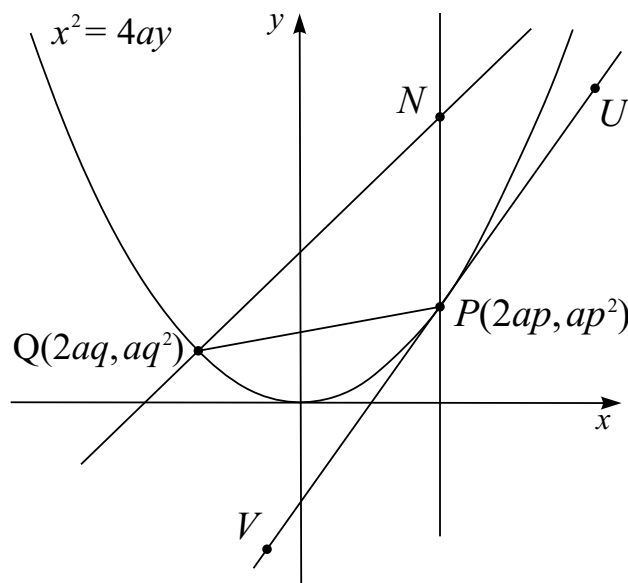
2

(ii) Find the first time that the particle is the furthest from the origin.

3

QUESTION THIRTEEN (Continued)

(d)



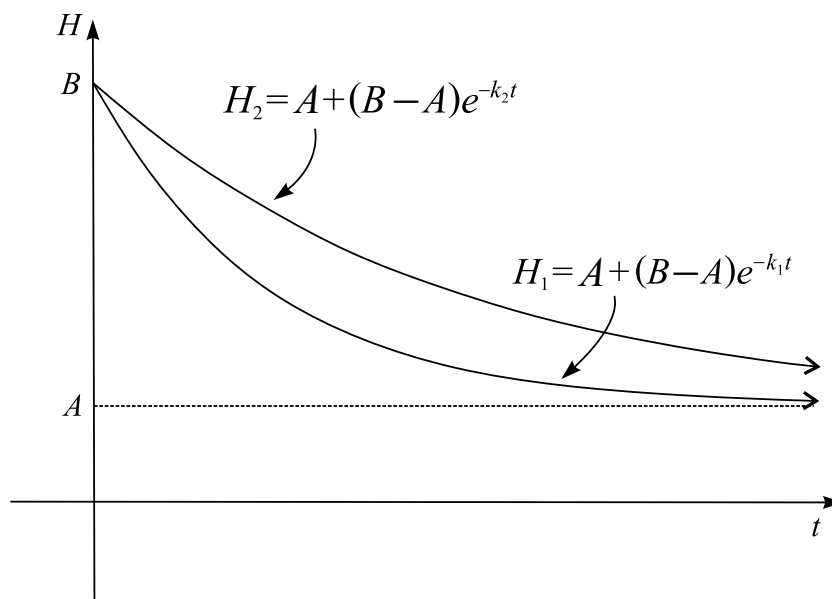
The diagram above shows the parabola $x^2 = 4ay$ with a tangent at the point $P(2ap, ap^2)$ and a normal at the point $Q(2aq, aq^2)$. The vertical line through P has been constructed, and it intersects the normal through Q at N . The points U and V lie on the tangent to the parabola at P , as shown.

The chord PQ has equation $x(p + q) - 2y - 2apq = 0$. (Do NOT prove this.)

- (i) Find the gradient of the tangent to the parabola at P . 1
- (ii) Show that if PQ is a focal chord, then the normal to the parabola at Q is parallel to the tangent to the parabola at P . 1
- (iii) The circle through the points N , Q and P is constructed. Use the reflection property of the parabola (that is, that $\angle QPV = \angle NPU$) to show that if PQ is a focal chord, the parabola and the circle have a common tangent at P . 2

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



A liquid is heated to B degrees Celsius before being poured into two different containers, container one and container two. The containers are then placed in a room that is A degrees Celsius, where $A < B$. The temperature H of each liquid after t minutes is graphed above. The temperatures of the liquids in container one and container two after t minutes are given by

$$H_1 = A + (B - A)e^{-k_1 t}$$

$$H_2 = A + (B - A)e^{-k_2 t}$$

respectively, where k_1 and k_2 are positive constants.

It is known that it takes the liquid in container two twice as long to cool to a given temperature as it does for the liquid in container one.

(i) Show that $k_1 = 2k_2$.

2

(ii) Find the largest difference in temperatures between the liquids in the two containers.

3

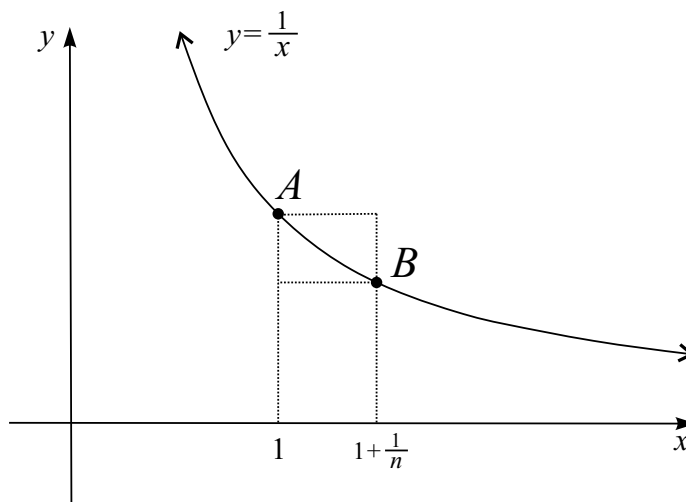
QUESTION FOURTEEN (Continued)

- (b) Use mathematical induction to show that, for any integer
- $n \geq 0$
- ,

3

$$\lim_{p \rightarrow \infty} \frac{p!}{(p-n)!p^n} = 1.$$

- (c)



The diagram above shows the points A and B on the curve $y = \frac{1}{x}$ with x -coordinates 1 and $(1 + \frac{1}{n})$ respectively, where $n > 0$.

- (i) By referring to the diagram, explain why

2

$$\frac{1}{n+1} \leq \int_1^{1+\frac{1}{n}} \frac{dx}{x} \leq \frac{1}{n}.$$

- (ii) Prove that

2

$$e^{\frac{n}{n+1}} \leq (1 + \frac{1}{n})^n \leq e.$$

- (iii) Deduce that
- $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$
- .

1

- (d) Let
- $a_{n,k}$
- be the term containing
- x^k
- in the expansion of
- $(1+x)^n$
- , where
- $0 < x < 1$
- .

2

It is known that $\lim_{n \rightarrow \infty} \sum_{k=0}^n a_{n,k} = \sum_{k=0}^{\infty} \lim_{n \rightarrow \infty} a_{n,k}$.

Use your answers to (b) and (c)(iii) to show that

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots.$$

————— End of Section II —————

END OF EXAMINATION

Extension 1 Maths - Trial 2018

Multiple Choice

① $A(-4, 1)$, $B(2, 7)$
 $S: 1$

$$x = \frac{-4 + 10}{6}$$

$$= 1$$

$$y = \frac{1 + 35}{6}$$

$$= 6$$

$\Rightarrow B$

② C

③ $3x + y + 2 = 0$ $y = 2 + x$
 $m_1 = -3$ $m_2 = 1$

$$\tan \theta = \left| \frac{-3 - 1}{1 - (-3) \times 1} \right|$$

$$= \left| \frac{-4}{2} \right|$$

$$\theta = \tan^{-1} 2$$

$$\div 63^\circ \quad \Rightarrow D$$

④ $\frac{d}{dx} (\sin^{-1} 2x) = \frac{1}{\sqrt{1 - (2x)^2}} \times 2$

$$= \frac{2}{\sqrt{1 - 4x^2}} \quad \Rightarrow A$$

⑤ $N = \sqrt{6x - x^3}$

$$\frac{1}{2} v^2 = \frac{1}{2} (6x - x^3)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{1}{2} (6 - 3x^2)$$

When $x = 2$: $\ddot{x} = \frac{1}{2} (6 - 3 \times 2^2)$

$$= -3 \text{ ms}^{-2} \quad \Rightarrow D$$

$$\textcircled{6} \int_0^k \frac{dx}{9+x^2} = \left[\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) \right]_0^k$$

$$= \frac{1}{3} \left(\tan^{-1} \frac{k}{3} - 0 \right)$$

$$\frac{1}{3} \tan^{-1} \frac{k}{3} = \frac{\pi}{9}$$

$$\tan^{-1} \frac{k}{3} = \frac{\pi}{3}$$

$$\frac{k}{3} = \tan \frac{\pi}{3}$$

$$= \sqrt{3}$$

$$\therefore k = 3\sqrt{3}$$

$$\Rightarrow C$$

$$\textcircled{7} x = (y+2)^2 + 1$$

$$(y+2)^2 = x-1$$

$$y+2 = -\sqrt{x-1}$$

$$y = -2 - \sqrt{x-1}$$

$$\Rightarrow C$$

since range will be $y \leq -2$

$$\textcircled{8} x = 6t, \quad y = -3t^2 + 3$$

$$t = \frac{x}{6} \quad \therefore y = -3\left(\frac{x}{6}\right)^2 + 3$$

$$y = -\frac{x^2}{12} + 3$$

$$12y = -x^2 + 36$$

$$x^2 = -12(y-3)$$

$\therefore$ Focal length = 3, Shifted up by 3, concave down.

$\therefore$ Focus is $(0,0) \Rightarrow B$

Alternatively: $x^2 = 4ay : (2at, at^2)$
 $a = 3$ gives $(6t, 3t^2)$

So $(6t, -3t^2 + 3)$ is the standard parabola with vertex $(0,0)$ and focus $(0,3)$ reflected in y-axis & shifted up by 3.

⑨ Reflex $\angle AOC = 2\theta$ (angle at centre = $2 \times$ angle at circumference)

$\therefore \angle AOC = 360^\circ - 2\theta$ (angles in a revolution)

$\angle OAT = \angle OCT = 90^\circ$ (angle between radius and tangent)

By angle sum of $ATCO$:

$$\angle ATC + 90^\circ + 90^\circ + 360^\circ - 2\theta = 360^\circ$$

$$\angle ATC = 2\theta - 180^\circ$$

$\Rightarrow A$

⑩ $P(2) = 0$

$$\therefore 16a + 16b + 16c + 16d + 16e = 0$$

$$a + b + c + d + e = 0 \quad (1)$$

$P(-2) = 32$:

$$16a - 16b + 16c - 16d + 16e = 32$$

$$a - b + c - d + e = 2 \quad (2)$$

① + ②:

$$2a + 2c + 2e = 2$$

$$a + c + e = 1$$

sub. into ①: $b + d + (a + c + e) = 0$

$$\therefore b + d = -1$$

$\Rightarrow B$

Question 11

(a) $(2x - y)(4x^2 + 2xy + y^2)$ ✓

(b) $\int \frac{dx}{\sqrt{9-x^2}} = \sin^{-1}\left(\frac{x}{3}\right) + c$ ✓

(c) $\int \cos^2 x \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx$ ✓
 $= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right] + c$ ✓
 $= \frac{1}{2} x + \frac{1}{4} \sin 2x + c$

(d) (i) $\alpha + \beta + \gamma = -\frac{18}{3}$

$= 6$

$\alpha\beta\gamma = -\frac{15}{3}$

$= -5$

} ✓

(ii) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$ ✓
 $= -\frac{6}{5}$

(e) $\frac{6}{x+2} \leq 1 \quad x \neq -2$

$6(x+2) \leq (x+2)^2$ ✓

$(x+2)^2 - 6(x+2) \geq 0$

$(x+2)[(x+2)-6] \geq 0$

$(x+2)(x-4) \geq 0$ ✓



$\therefore x < -2, x \geq 4$ ✓

✓ (obtaining -2 and 4)

$$(f) \int_0^{\frac{\pi}{2}} \cos x (1 - \sin x)^2 dx$$

$$= - \int_0^{\frac{\pi}{2}} -\cos x (1 - \sin x)^2 dx$$

$$= - \int_1^0 u^2 du \quad \checkmark$$

$$= \int_0^1 u^2 du$$

$$= \left[\frac{u^3}{3} \right]_0^1$$

$$= \frac{1}{3} \quad \checkmark$$

$$u = 1 - \sin x$$

$$du = -\cos x dx$$

$$x = 0, u = 1$$

$$x = \frac{\pi}{2}, u = 0 \quad \checkmark$$

$$(g) \text{ General term: } {}^9C_r (3x^2)^r \left(-\frac{2}{x}\right)^{9-r} \quad \checkmark$$

$$= {}^9C_r 3^r (-2)^{9-r} x^{2r} x^{r-9}$$

$$= {}^9C_r 3^r (-2)^{9-r} x^{3r-9}$$

$$\text{Constant when } 3r - 9 = 0$$

$$r = 3 \quad \checkmark$$

$$\therefore \text{ Constant term is } {}^9C_3 3^3 (-2)^6 \quad \checkmark$$

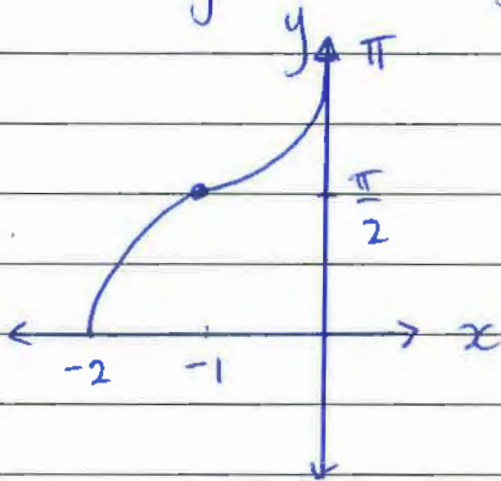
$$= 145152$$

Question 12

(a)(i) Domain: $-2 \leq x \leq 0$ ✓

Range: $0 \leq y \leq \pi$ ✓

(ii)



(b)(i) $\operatorname{cosec} 2\alpha - \cot 2\alpha = \tan \alpha$

LHS = $\operatorname{cosec} 2\alpha - \cot 2\alpha$

$$= \frac{1}{\sin 2\alpha} - \frac{1}{\tan 2\alpha}$$

$$= \frac{1}{2\sin \alpha \cos \alpha} - \frac{1 - \tan^2 \alpha}{2\tan \alpha}$$
 ✓

$$= \frac{1}{2\sin \alpha \cos \alpha} - \frac{(1 - \tan^2 \alpha) \cos^2 \alpha}{2\sin \alpha \cos \alpha}$$

$$= \frac{1 - (\cos^2 \alpha - \sin^2 \alpha)}{2\sin \alpha \cos \alpha}$$

$$= \frac{1 - \cos^2 \alpha + \sin^2 \alpha}{2\sin \alpha \cos \alpha}$$

$$= \frac{2\sin^2 \alpha}{2\sin \alpha \cos \alpha}$$

$$= \tan \alpha$$
 ✓

$$= \text{RHS}$$

Alternatively: Let $t = \tan \alpha$

$$\text{then } \sin 2\alpha = \frac{2t}{1+t^2}, \quad \tan \alpha = \frac{2t}{1-t^2}$$

$$\begin{aligned} \therefore \operatorname{cosec} 2\alpha - \cot 2\alpha &= \frac{1+t^2}{2t} - \frac{1-t^2}{2t} \\ &= \frac{2t^2}{2t} \end{aligned}$$

$$= t$$

$$= \tan \alpha$$

$$(ii) \quad \tan 15^\circ = \operatorname{cosec} 30^\circ - \cot 30^\circ$$

$$= \frac{1}{\sin 30^\circ} - \frac{1}{\tan 30^\circ}$$

$$= 2 - \sqrt{3}$$

$$(c) (i) \quad f(2) = \tan\left(\frac{2}{2}\right) - 2$$

$$\doteq -0.44 \dots < 0$$

$$f(3) = \tan\left(\frac{3}{2}\right) - 3$$

$$\doteq 11.10 \dots > 0$$

Since $f(x)$ changes sign between $x=2$ and $x=3$,
and $f(x)$ is continuous over $2 \leq x \leq 3$

$$(ii) \quad f(x) = \tan\left(\frac{x}{2}\right) - x$$

$$f'(x) = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) - 1$$

$$x_2 = 2.5 - \frac{\tan\left(\frac{2.5}{2}\right) - 2.5}{\frac{1}{2\cos^2\left(\frac{2.5}{2}\right)} - 1}$$

$$\doteq 2.37 \quad (2 \text{ decimal places})$$

(d) $\angle DAC = \angle DBC$ (angles on circumference subtended by arc DC) ✓

$\angle BAC = \angle BCX$ (angle between tangent and chord = angle in alternate segment)

Since $\angle DAC = \angle BAC$,
 $\angle DBC = \angle BCX$

$\therefore DB \parallel YX$ (alternate angles are equal) ✓

(e) (i) $\tan 60^\circ = \frac{\frac{1}{2}x}{h}$

$$\sqrt{3} = \frac{\frac{1}{2}x}{h}$$

$$x = 2h\sqrt{3}$$
 ✓

(ii) $V = \frac{1}{3}x^2h$

$$= \frac{1}{3}(2h\sqrt{3})^2 \times h$$

$$V = 4h^3$$

$$h^3 = \frac{1}{4}V$$

$$h = \left(\frac{1}{4}V\right)^{\frac{1}{3}}$$
 ✓

(ii) $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$

$$\frac{dV}{dt} = 0.5$$

$$\frac{dh}{dV} = \frac{1}{3} \left(\frac{1}{4}V\right)^{-\frac{2}{3}} \times \frac{1}{4}$$

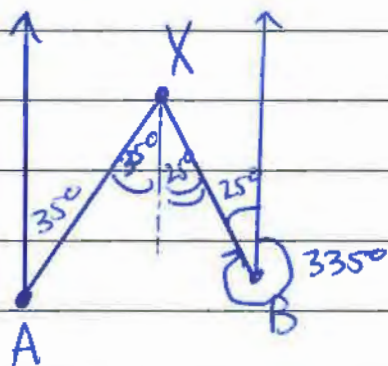
$$= \frac{1}{12 \left(\frac{1}{4}V\right)^{\frac{2}{3}}}$$
 ✓

After 8 mins, $V = 8 \times 0.5 = 4$

$$\therefore \frac{dh}{dt} = \frac{1}{12(1)^{\frac{2}{3}}} \times 0.5 = \frac{1}{24} \text{ m/min}$$
 ✓

Question 13

(a) (i)



$$\angle AXB = 35^\circ + 25^\circ \\ = 60^\circ$$

$$(ii) \tan 18^\circ = \frac{200}{AX}$$

$$AX = \frac{200}{\tan 18^\circ}$$

$$\text{Similarly } BX = \frac{200}{\tan 15^\circ}$$

✓
} ✓ (both)

By cosine rule:

$$AB^2 = \left(\frac{200}{\tan 18^\circ}\right)^2 + \left(\frac{200}{\tan 15^\circ}\right)^2 - 2 \times \frac{200}{\tan 18^\circ} \times \frac{200}{\tan 15^\circ} \times \cos 60^\circ \\ = 476570.71...$$

$$AB \div 690.34$$

$$\approx 690 \text{ m (nearest metre)}$$

(b) Let the roots be $\frac{x}{r}, x, xr$

$$\text{Product of roots: } \frac{x}{r} \times x \times xr = -\frac{-216}{1}$$

$$x^3 = 216$$

$$x = 6$$

$$\text{Sum of roots: } \frac{6}{r} + 6 + 6r = -\frac{9}{1}$$

$$\frac{2}{r} + 2 + 2r = -3$$

$$2r^2 + 5r + 2 = 0$$

$$(2r+1)(r+2) = 0$$

$$r = -\frac{1}{2} \text{ or } r = -2$$

$$\therefore \text{ roots are } -3, 6, -12 \quad \checkmark$$

Alternatively:

Let the roots be $\alpha, \alpha r, \alpha r^2$

$$\text{Product of roots: } \alpha \cdot \alpha r \cdot \alpha r^2 = 216$$

$$\alpha^3 r^3 = 216$$

$$\alpha r = 6 \quad \checkmark$$

$$\text{Sum of roots: } \alpha + \alpha r + \alpha r^2 = -9$$

$$\alpha + 6 + 6r = -9$$

$$\alpha + 6r = -15 \quad (*)$$

$$\text{Two at a time: } \alpha \cdot \alpha r + \alpha \cdot \alpha r^2 + \alpha r \cdot \alpha r^2 = k$$

$$\alpha^2 r + \alpha^2 r^2 + \alpha^2 r^3 = k$$

$$\alpha r = 6: \quad 6\alpha + 36 + 36r = k$$

$$6(\alpha + 6r) + 36 = k$$

$$6x - 15 + 36 = k \quad \checkmark \quad (\text{from } *)$$

$$k = -54 \quad \checkmark$$

$\therefore$ Polynomial is $x^3 + 9x^2 - 54x - 216 = 0$ with 6 as one root.

$$(x-6)(x^2 + 15x + 36) = 0$$

$$(x-6)(x+3)(x+12) = 0$$

$$\therefore \text{ roots are } x = -3, x = 6, x = -12 \quad \checkmark$$

$$\begin{aligned}
 \text{(c) (i)} \quad x &= \sqrt{3} \sin 2t - \cos 2t - 1 \\
 \dot{x} &= 2\sqrt{3} \cos 2t + 2 \sin 2t \\
 \ddot{x} &= -4\sqrt{3} \sin 2t + 4 \cos 2t \\
 &= -4(\sqrt{3} \sin 2t - \cos 2t) \\
 &= -4(x+1)
 \end{aligned}$$

Alternatively:

$$\begin{aligned}
 \text{let } \sqrt{3} \sin 2t - \cos 2t &= R \sin(2t + \alpha) \\
 &= R \sin 2t \cos \alpha + R \cos 2t \sin \alpha
 \end{aligned}$$

$$\Rightarrow R \cos \alpha = \sqrt{3} \quad \text{①}$$

$$R \sin \alpha = -1 \quad \text{②}$$

$$\begin{aligned}
 \text{①}^2 + \text{②}^2: \quad R^2 \cos^2 \alpha + R^2 \sin^2 \alpha &= (\sqrt{3})^2 + (-1)^2 \\
 R^2 &= 4
 \end{aligned}$$

$$R = 2 \quad \text{taking } R > 0.$$

$$\left. \begin{aligned} 2 \cos \alpha &= \sqrt{3} \\ 2 \sin \alpha &= -1 \end{aligned} \right\} \alpha \text{ 4th quadrant.}$$

$$\sin \alpha = -\frac{1}{2}$$

$$\alpha = -\frac{\pi}{6} \quad \left(\text{or } \frac{11\pi}{6}, \dots \right)$$

$$\therefore x = 2 \sin\left(2t - \frac{\pi}{6}\right) - 1$$

$$\dot{x} = 4 \cos\left(2t - \frac{\pi}{6}\right)$$

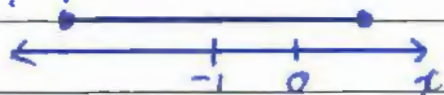
$$\ddot{x} = -8 \sin\left(2t - \frac{\pi}{6}\right)$$

$$= -4 \times 2 \sin\left(2t - \frac{\pi}{6}\right)$$

$$= -4(x+1)$$

(ii) Centre of motion: $x = -1$

Furthest from the origin will be in the negative direction,
when $\dot{x} = 0$: ↓



When $t = 0$, $x = +2\sqrt{3}$

i.e. particle is initially moving in positive direction, so
will be "furthest from the origin" the second time that
it comes to rest.

$$\dot{x} = 0: 2\sqrt{3} \cos 2t + 2 \sin 2t = 0$$

$$\tan 2t = -\sqrt{3}$$

$$2t = \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, \dots$$

$$= \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{8\pi}{3}, \dots$$

$$t = \frac{\pi}{3}, \frac{5\pi}{6}, \frac{4\pi}{3}, \dots$$

$\therefore$ particle is furthest from the origin when $t = \frac{5\pi}{6}$

$$(d)(i) \quad y = \frac{x^2}{4a}$$

$$y' = \frac{x}{2a}$$

when $x = 2ap$, $y' = p$ ✓

(ii) Similarly, at $Q(2aq, aq^2)$, $y' = q$
 $\therefore$ m normal at $Q = -\frac{1}{q}$

If PQ is a focal chord, $(0, a)$ lies on $x(p+q) - 2y - 2apq = 0$
 $\therefore -2a - 2apq = 0$
 $pq = -1$
 $p = -\frac{1}{q}$ ✓

$\therefore$ tangent at P is parallel to normal at Q .

(iii) From (ii), $QN \parallel VU$

$\therefore \angle NPV = \angle QNP$ (alternate angles, $QN \parallel VU$) ✓

but $\angle QPV = \angle NPV$ (reflection property)

$\therefore \angle QPV = \angle QNP$

$\therefore UV$ is a tangent to circle QPN (converse of alternate segment theorem) ✓

Question 14

(a) (i) When $t = t_0$, H_1 will equal H_2 when $t = 2t_0$

$$\text{i.e. } A + (B-A)e^{-k_2 \times 2t_0} = A + (B-A)e^{-k_1 t_0} \quad \checkmark$$

$$e^{-2k_2 t_0} = e^{-k_1 t_0}$$

$$-2k_2 t_0 = -k_1 t_0$$

$$k_1 = 2k_2 \quad \checkmark$$

(ii) Difference in temperatures = $H_2 - H_1$

$$\begin{aligned} &= A + (B-A)e^{-k_2 t} - (A + (B-A)e^{-k_1 t}) \\ &= (B-A)(e^{-k_2 t} - e^{-k_1 t}) \\ &= (B-A)(e^{-k_2 t} - e^{-2k_2 t}) \quad \checkmark \end{aligned}$$

Method 1: Let $M = H_2 - H_1$

$$\frac{dM}{dt} = (B-A)(-k_2 e^{-k_2 t} + 2k_2 e^{-2k_2 t})$$

$$\text{Let } \frac{dM}{dt} = 0:$$

$$-k_2 e^{-k_2 t} + 2k_2 e^{-2k_2 t} = 0$$

$$\frac{2}{e^{2k_2 t}} = \frac{1}{e^{k_2 t}}$$

$$2 = e^{k_2 t}$$

$$k_2 t = \ln 2$$

$$t = \frac{1}{k_2} \ln 2 \quad \checkmark$$

$$\frac{d^2 M}{dt^2} = (B-A) \left(k_2^2 e^{-k_2 t} - 4k_2^2 e^{-2k_2 t} \right)$$

When $t = \frac{1}{k_2} \ln 2$:

$$\frac{d^2 M}{dt^2} = (B-A) \times k_2^2 \left(e^{-k_2 \cdot \frac{1}{k_2} \ln 2} - 4e^{-2k_2 \cdot \frac{1}{k_2} \ln 2} \right)$$

$$= (B-A) \cdot k_2^2 (e^{-\ln 2} - 4e^{-2\ln 2})$$

$$= (B-A) \cdot k_2^2 \left(\frac{1}{2} - \frac{1}{4} \right)$$

$$= (B-A) \cdot k_2^2 x - \frac{1}{2}$$

< 0 since $B > A$ and $k_2^2 > 0$.

$\therefore$ Max. occurs when $t = \frac{1}{k_2} \ln 2$.

$$\begin{aligned} \therefore \text{Max. difference} &= (B-A) \left(e^{-k_2 \times \frac{1}{k_2} \ln 2} - e^{-2k_2 \times \frac{1}{k_2} \ln 2} \right) \\ &= (B-A) \left(\frac{1}{2} - \frac{1}{4} \right) \\ &= \frac{B-A}{4} \text{ } ^\circ\text{C} \quad \checkmark \end{aligned}$$

Method 2:

$$H_2 - H_1 = (B-A)(e^{-k_2 t} - e^{-2k_2 t})$$

let $x = e^{-k_2 t}$

$$H_2 - H_1 = (B-A)(x - x^2)$$

$B-A$ is constant, and $y = x - x^2$ represents a concave down parabola, with local max. at

$$\begin{aligned} x &= -\frac{1}{2x-1} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} y &= \frac{1}{2} - \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{2} - \frac{1}{4} \\ &= \frac{1}{4} \end{aligned} \quad \checkmark$$

$$\begin{aligned} \text{Check: } x = \frac{1}{2} \text{ gives } e^{-k_2 t} &= \frac{1}{2} \\ -k_2 t &= \ln\left(\frac{1}{2}\right) \\ &= -\ln 2 \end{aligned}$$

$t = \frac{1}{k_2} \ln 2$ i.e. a valid result for t .

$$\begin{aligned} \therefore \text{Max. difference} &= (B-A) \times \frac{1}{4} \\ &= \frac{B-A}{4} \text{ } ^\circ\text{C} \quad \checkmark \end{aligned}$$

(b) Step 1: $n=0$

$$\begin{aligned}\text{LHS} &= \lim_{p \rightarrow \infty} \frac{p!}{(p-0)! p^0} \\ &= \lim_{p \rightarrow \infty} \frac{p!}{p!} \\ &= \lim_{p \rightarrow \infty} 1 \\ &= 1 \\ &= \text{RHS}\end{aligned}$$

$\therefore$ true for $n=0$.

Step 2: Assume true for $n=k$.

i.e. $\lim_{p \rightarrow \infty} \frac{p!}{(p-k)! p^k} = 1$

Step 3: Prove true for $n=k+1$

To prove: $\lim_{p \rightarrow \infty} \frac{p!}{(p-(k+1))! p^{k+1}} = 1$

$$\lim_{p \rightarrow \infty} \frac{p!}{(p-k-1)! p^{k+1}} = 1$$

$$\begin{aligned}\text{LHS} &= \lim_{p \rightarrow \infty} \frac{p!}{(p-k-1)! p^{k+1}} \\ &= \lim_{p \rightarrow \infty} \frac{p! (p-k)}{(p-k)! p^{k+1}}\end{aligned}$$

$$= \lim_{p \rightarrow \infty} \left[\frac{p!}{(p-k)! p^k} \times \frac{p-k}{p} \right]$$

$$= \lim_{p \rightarrow \infty} \frac{p!}{(p-k)! p^k} \times \lim_{p \rightarrow \infty} \frac{p-k}{p} \quad \text{since both limits are defined.}$$

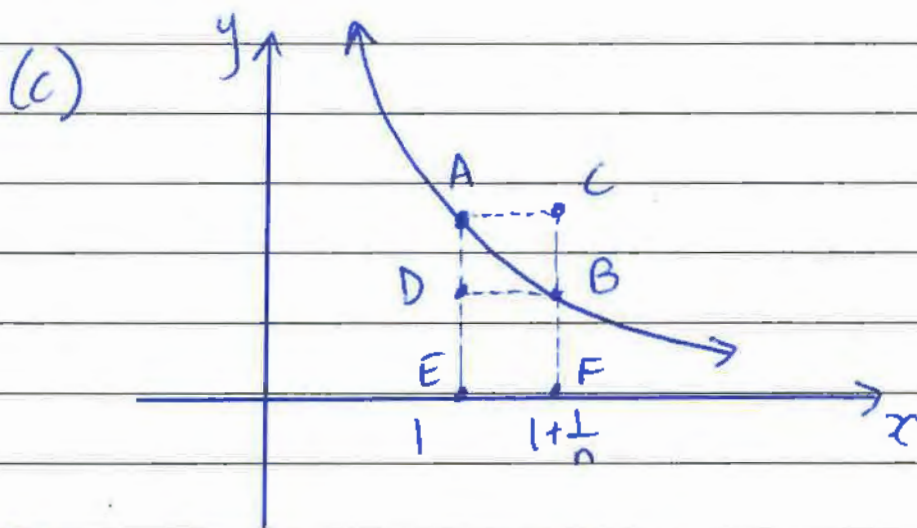
$$= 1 \times \lim_{p \rightarrow \infty} \left(1 - \frac{k}{p} \right) \quad (\text{from assumption})$$

$$= 1 \times (1 - 0)$$

$$= 1$$

$$= \text{RHS}$$

$\therefore$ true for $n=k+1$, so true for integers $n \geq 0$ by mathematical induction.



(i) Area DBFE $\leq$ Area under curve $\leq$ Area AEFC ✓

$$\frac{1}{n} \times \frac{1}{1 + \frac{1}{n}} \leq \int_1^{1 + \frac{1}{n}} \frac{dx}{x} \leq \frac{1}{n} \times 1$$

$$\therefore \frac{1}{n+1} \leq \int_1^{1 + \frac{1}{n}} \frac{dx}{x} \leq \frac{1}{n}$$
 ✓

(ii) $\int_1^{1 + \frac{1}{n}} \frac{dx}{x} = [\ln x]_1^{1 + \frac{1}{n}}$
 $= \ln(1 + \frac{1}{n}) - \ln(1)$
 $= \ln(1 + \frac{1}{n})$ ✓

$$\therefore \frac{1}{n+1} \leq \ln(1 + \frac{1}{n}) \leq \frac{1}{n}$$

$$e^{\frac{1}{n+1}} \leq e^{\ln(1 + \frac{1}{n})} \leq e^{\frac{1}{n}}$$

$$e^{\frac{n}{n+1}} \leq (1 + \frac{1}{n})^n \leq e$$
 ✓

(iii) $\lim_{n \rightarrow \infty} e^{\frac{n}{n+1}} = \lim_{n \rightarrow \infty} e^{\frac{1}{1 + \frac{1}{n}}}$
 $= e^{\frac{1}{1+0}}$
 $= e$

Since $e^{\frac{n}{n+1}} \rightarrow e$ from below, $(1 + \frac{1}{n})^n \rightarrow e$ also.

i.e. $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ ✓

$$(d) \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \quad \checkmark \quad \left(\text{recognising application of binomial theorem to } \left(1 + \frac{1}{n}\right)^n \text{ and writing correct expansion}\right)$$

$$= \sum_{k=0}^{\infty} \lim_{n \rightarrow \infty} \binom{n}{k} \frac{1}{n^k} \quad \text{since } 0 < \frac{1}{n} < 1 \text{ for } n > 0.$$

$$\therefore \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \lim_{n \rightarrow \infty} \frac{n!}{(n-k)! k! n^k}$$

$$= \sum_{k=0}^{\infty} \lim_{n \rightarrow \infty} \frac{n!}{(n-k)! n^k} \times \frac{1}{k!}$$

From (b): $\lim_{p \rightarrow \infty} \frac{p!}{(p-n)! p^n} = 1$

Using dummy variables, $p \Rightarrow n$, $n \Rightarrow k$

$$\lim_{n \rightarrow \infty} \frac{n!}{(n-k)! n^k} = 1$$

$$\therefore \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} 1 \times \frac{1}{k!}$$

Applying result from (c)(iii):

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

$$= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \quad \checkmark$$